Quick select

Rank
Def: Given a list, the rank r element is the element that will be at index r in the sorted version.

[3,1,2]
rank| 2 0 1

rank 0: smallest element
rank len: largest element

Quick Select
- find the rank-i element in a list

rank 0 or len can be done in O(n) time
but rank i takes longer

the naive approach is to sort and look at index rank

steps:
- partition the list randomly

0 1 2 3 4 5 6 7 8 9
↓
[16, 52, 13, 12, 19, 80, 1, 32, 5, 25]

random pivot 19
valid partition

[16, 13, 12, 1, 5, 19, 52, 80, 32, 25]
pivot is at index 5 = rank

- pivot is at rank position
- target rank is on one side, depending on if i is < or > r

- depends on how lucky you are with the pivot

O(n)